sparsity

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1 Introduction

A variational quantum circuit (VQC) is a parameterized quantum circuit whose trainable parameters are optimized by a classical optimizer. Let

$$U(\boldsymbol{\theta}) = U_p(\theta_p)U_{p-1}(\theta_{p-1})\cdots U_1(\theta_1) \tag{1}$$

be a parameterized circuit with $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p) \in \mathbb{R}^p$. Starting from an initial state ρ_0, the circuit prepares

$$\rho(\boldsymbol{\theta}) = U(\boldsymbol{\theta})\rho_0 U^\dagger(\boldsymbol{\theta}). \tag{2}$$

Given an observable H, the objective function is defined as

$$C(\boldsymbol{\theta}) = \mathrm{Tr}\left[H\rho(\boldsymbol{\theta})\right]. \tag{3}$$

For a pure initial state $|\psi_0\rangle$, this can equivalently be written as

$$C(\boldsymbol{\theta}) = \langle\psi_0| U^\dagger(\boldsymbol{\theta}) H U(\boldsymbol{\theta}) |\psi_0\rangle . \tag{4}$$

In many VQCs, the parameterized gates are generated by Hermitian operators. A common form is

$$U_j(\theta_j) = e^{-i\theta_j G_j/2}, \tag{5}$$

where G_j is the generator of the j-th parameterized gate. For Pauli rotation gates, G_j is a Pauli string.

The gradient of the objective function is

$$\nabla C(\boldsymbol{\theta}) = \left(\partial_1 C(\boldsymbol{\theta}), \partial_2 C(\boldsymbol{\theta}), \ldots, \partial_p C(\boldsymbol{\theta})\right), \tag{6}$$

where

$$\partial_j C(\boldsymbol{\theta}) = \frac{\partial C(\boldsymbol{\theta})}{\partial \theta_j}. \tag{7}$$

For gates whose generators have two distinct eigenvalues, the parameter-shift rule gives an exact expression for each partial derivative:

$$\partial_j C(\boldsymbol{\theta}) = \frac{1}{2}\left[C\left(\boldsymbol{\theta} + \frac{\pi}{2}\mathbf{e}_j\right) - C\left(\boldsymbol{\theta} - \frac{\pi}{2}\mathbf{e}_j\right)\right], \tag{8}$$

where $\mathbf{e}_j$ is the unit vector in the j-th parameter direction. Thus, computing the full gradient by the parameter-shift rule requires $2p$ circuit evaluations.

In contrast, simultaneous perturbation stochastic approximation (SPSA) estimates the whole gradient using only two circuit evaluations per perturbation sample:

$$\hat{g}(\boldsymbol{\theta}) = \frac{C(\boldsymbol{\theta} + c\boldsymbol{\Delta}) - C(\boldsymbol{\theta} - c\boldsymbol{\Delta})}{2c} \boldsymbol{\Delta}^{-1}, \tag{9}$$

where $c > 0$ is the perturbation scale and each component of $\boldsymbol{\Delta}$ is usually sampled from $\{-1, 1\}$. SPSA is circuit-efficient, but its gradient estimate is stochastic and generally less accurate than the parameter-shift rule.

This motivates selective gradient estimation methods for VQCs whose gradients are sparse. In such cases, only a small number of parameters contribute significantly to the objective, so estimating all partial derivatives uniformly is inefficient.

1.1 Sparse Parameter Gradients

We now introduce the sparse-gradient assumption considered in this work. Instead of requiring most gradient components to be exactly zero, we measure sparsity by the concentration of the squared gradient magnitude.

Let

$$\mathbf{g}(\boldsymbol{\theta}) = \nabla C(\boldsymbol{\theta}) \in \mathbb{R}^p \tag{10}$$

be the gradient of a p-parameter VQC. For an integer k, let $S_k(\boldsymbol{\theta}) \subseteq \{1, \dots, p\}$ denote the index set of the k largest components of $\mathbf{g}(\boldsymbol{\theta})$ in absolute value:

$$S_k(\boldsymbol{\theta}) = \mathrm{TopK}\left(\{|\partial_j C(\boldsymbol{\theta})|\}_{j=1}^{p}\right). \tag{11}$$

Definition 1 ((k, ρ)-sparse parameter gradient)**.** *The gradient $\nabla C(\boldsymbol{\theta})$ is called (k, ρ)-sparse if k is much smaller than the total number of parameters,*

$$k \ll p, \tag{12}$$

and there exists $\rho \in (0, 1]$ such that

$$\frac{\sum_{j \in S_k(\boldsymbol{\theta})} \left(\partial_j C(\boldsymbol{\theta})\right)^2}{\sum_{j=1}^{p} \left(\partial_j C(\boldsymbol{\theta})\right)^2} \geq \rho. \tag{13}$$

Equivalently, the top-k gradient components contain at least a ρ fraction of the total squared gradient magnitude:

$$\|\mathbf{g}_{S_k}(\boldsymbol{\theta})\|_2^2 \geq \rho \, \|\mathbf{g}(\boldsymbol{\theta})\|_2^2, \tag{14}$$

where $\mathbf{g}_{S_k}$ denotes the vector obtained by keeping only the entries indexed by S_k and setting all other entries to zero.

The condition $k \ll p$ is essential. It means that the dominant gradient information is concentrated in a subset of parameters whose size is much smaller than the total number of trainable parameters. For a sequence of circuits with increasing parameter dimension p, this can be expressed more formally as $k = o(p)$.

When ρ is close to one, most of the gradient energy is concentrated in the top-k components. In the special case $\rho = 1$, the gradient is exactly k-sparse. This definition is more suitable for practical VQCs than exact support sparsity, because finite-shot noise and numerical fluctuations may make inactive components appear nonzero.

Under this assumption, the parameter-shift rule remains accurate but is inefficient because it estimates all p partial derivatives, including many components that contribute little to the total gradient energy. SPSA is circuit-efficient because it perturbs all parameters Ssimultaneously, but its stochastic estimate may be inaccurate on the important active components. Select-SPSA is designed for this (k, ρ)-sparse regime: it aims to identify and estimate the dominant gradient components more accurately while using fewer circuit evaluations than the full parameter-shift rule.

1.2 Examples of Gradient Sparsity

First, we explain how exact sparse parameter gradients can arise from the structure of a variational quantum circuit. Throughout this section, we use O to denote the measured observable, or equivalently the Hamiltonian defining the objective function.

Consider an n-qubit variational quantum circuit

$$U(\boldsymbol{\theta}) = U_p(\theta_p)U_{p-1}(\theta_{p-1}) \cdots U_1(\theta_1), \tag{15}$$

where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)$ is the vector of trainable parameters. Starting from an initial state ρ_0, the objective function is

$$C(\boldsymbol{\theta}) = \mathrm{Tr}\left[OU(\boldsymbol{\theta})\rho_0 U^\dagger(\boldsymbol{\theta})\right]. \tag{16}$$

We assume that each parameterized gate has the form

$$U_j(\theta_j) = e^{-i\theta_j G_j/2}, \tag{17}$$

where G_j is a Hermitian generator. For Pauli rotation gates, G_j is a Pauli operator or a Pauli string.

For the j-th parameterized gate, decompose the circuit as

$$U(\boldsymbol{\theta}) = U_{>j}(\boldsymbol{\theta})U_j(\theta_j)U_{<j}(\boldsymbol{\theta}), \tag{18}$$

where $U_{<j}$ contains the gates before U_j and $U_{>j}$ contains the gates after U_j. Define the backward-evolved observable

$$O_j(\boldsymbol{\theta}) = U^\dagger_{>j}(\boldsymbol{\theta})OU_{>j}(\boldsymbol{\theta}). \tag{19}$$

Then the gradient component with respect to θ_j can be written as

$$\partial_j C(\boldsymbol{\theta}) = \frac{i}{2}\,\mathrm{Tr}\left[\rho_j(\boldsymbol{\theta})[G_j, O_j(\boldsymbol{\theta})]\right], \tag{20}$$

where

$$\rho_j(\boldsymbol{\theta}) = U_j(\theta_j) U_{<j}(\boldsymbol{\theta}) \rho_0 U_{<j}^\dagger(\boldsymbol{\theta}) U_j^\dagger(\theta_j). \tag{21}$$

Equation (20) is the basic tool for identifying inactive parameters. If the commutator $[G_j, O_j(\boldsymbol{\theta})]$ vanishes, then the corresponding gradient component is exactly zero.

Definition 2 (Exact sparse parameter gradients)**.** *Let*

$$\mathbf{g}(\boldsymbol{\theta}) = \nabla C(\boldsymbol{\theta}) = \big(\partial_1 C(\boldsymbol{\theta}), \ldots, \partial_p C(\boldsymbol{\theta})\big).$$

The gradient is called exactly k-sparse if, for some active set $\mathcal{A} \subseteq \{1, \ldots, p\}$ with $|\mathcal{A}| \leq k$ and $k = o(p)$,

$$\partial_j C(\boldsymbol{\theta}) = 0, \qquad \forall j \notin \mathcal{A}.$$

Equivalently, if $S_k(\boldsymbol{\theta})$ denotes the indices of the k largest entries of $\mathbf{g}(\boldsymbol{\theta})$ in absolute value, then

$$\frac{\sum_{j \in S_k(\boldsymbol{\theta})} \big(\partial_j C(\boldsymbol{\theta})\big)^2}{\sum_{j=1}^{p} \big(\partial_j C(\boldsymbol{\theta})\big)^2} = 1.$$

Thus, exact sparsity is the special case of (k, ρ)-sparsity with $\rho = 1$.

1.2.1 Causal-Cone-Induced Exact Sparsity

The first structural mechanism comes from locality and causal cones. It is based on the following logic:

local observable+local shallow circuit $\Rightarrow$ small backward causal cone $\Rightarrow$ many parameters outside the cone $\Rightarrow$

Suppose the observable is a sum of local terms,

$$O = \sum_{\alpha=1}^{M} c_\alpha O_\alpha, \tag{22}$$

where each O_α acts nontrivially only on a small subset of qubits. Then

$$C(\boldsymbol{\theta}) = \sum_{\alpha=1}^{M} c_\alpha C_\alpha(\boldsymbol{\theta}), \tag{23}$$

where

$$C_\alpha(\boldsymbol{\theta}) = \mathrm{Tr}\left[O_\alpha U(\boldsymbol{\theta}) \rho_0 U^\dagger(\boldsymbol{\theta})\right]. \tag{24}$$

For the j-th parameter, define

$$O_{\alpha,j}(\boldsymbol{\theta}) = U_{>j}^\dagger(\boldsymbol{\theta}) O_\alpha U_{>j}(\boldsymbol{\theta}). \tag{25}$$

Then

$$\partial_j C_\alpha(\boldsymbol{\theta}) = \frac{i}{2} \mathrm{Tr}\left[\rho_j(\boldsymbol{\theta}) [G_j, O_{\alpha,j}(\boldsymbol{\theta})]\right]. \tag{26}$$

The backward causal cone of O_α consists of all gates that cannot be eliminated from

$$U^\dagger(\boldsymbol{\theta}) O_\alpha U(\boldsymbol{\theta}). \tag{27}$$

If the gate generated by G_j lies outside this backward causal cone, then G_j has disjoint support from the backward-evolved observable $O_{\alpha,j}(\boldsymbol{\theta})$. Hence,

$$[G_j, O_{\alpha,j}(\boldsymbol{\theta})] = 0, \tag{28}$$

and therefore

$$\partial_j C_\alpha(\boldsymbol{\theta}) = 0. \tag{29}$$

Define the causal active set

$$\mathcal{B} = \{j \in \{1, \ldots, p\} : \exists \alpha \text{ such that } U_j(\theta_j) \text{ lies in the backward causal cone of } O_\alpha\}. \tag{30}$$

For every $j \notin \mathcal{B}$, all local contributions vanish:

$$\partial_j C_\alpha(\boldsymbol{\theta}) = 0, \qquad \forall \alpha. \tag{31}$$

Therefore,

$$\partial_j C(\boldsymbol{\theta}) = 0, \qquad \forall j \notin \mathcal{B}. \tag{32}$$

If

$$|\mathcal{B}| = k, \qquad k = o(p), \tag{33}$$

then the gradient is exactly sparse.

Concrete causal-cone example. Consider a five-qubit circuit with three layers of parameterized single-qubit rotations and two layers of local entangling gates:

$$U(\boldsymbol{\theta}) = U_R^{(3)} U_E^{(2)} U_R^{(2)} U_E^{(1)} U_R^{(1)}, \tag{34}$$

where

$$U_R^{(\ell)} = \prod_{i=1}^{5} R_y(\theta_i^{(\ell)}), \tag{35}$$

and

$$U_E^{(1)} = \mathrm{CNOT}_{2\to1}\,\mathrm{CNOT}_{4\to3}, \qquad U_E^{(2)} = \mathrm{CNOT}_{3\to2}\,\mathrm{CNOT}_{5\to4}\,. \tag{36}$$

Let the measured observable be the local operator

$$O = Z_1. \tag{37}$$

The corresponding circuit can be drawn as follows, where the shaded rotation gates are inside the backward causal cone of Z_1:

To see this, propagate Z_1 backwards through the circuit. The final rotation layer contributes only through the rotation on qubit 1. The entangling layer $U_E^{(2)}$ does not touch qubit 1, so the cone does not expand. The middle rotation

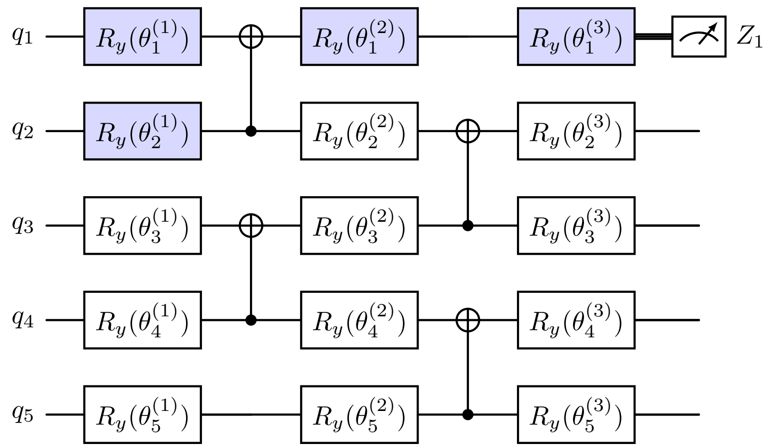

Figure 1: A five-qubit example of a backward causal cone. The observable is $O = Z_1$. Only the shaded parameterized gates can affect this local measurement.

layer again contributes only through qubit 1. When the observable is propagated through $\mathrm{CNOT}_{2\to1}$ in $U_E^{(1)}$, the support can expand from qubit 1 to qubits 1 and 2. Therefore, in the first rotation layer, only the rotations on qubits 1 and 2 can influence the measurement.

Thus, among the fifteen rotation parameters, only four parameters are in the backward causal cone:

$$\mathcal{B} = \left\{\theta_1^{(1)}, \theta_2^{(1)}, \theta_1^{(2)}, \theta_1^{(3)}\right\}. \tag{38}$$

All other rotation parameters lie outside the backward causal cone of Z_1 and have zero gradient:

$$\partial_{\theta_i^{(\ell)}} C(\boldsymbol{\theta}) = 0, \qquad \theta_i^{(\ell)} \notin \mathcal{B}. \tag{39}$$

This example illustrates the structural nature of causal-cone sparsity. The zero gradients do not require a special algebraic symmetry; they follow from the fact that local gates outside the backward causal cone cannot influence the local observable.

For a larger one-dimensional nearest-neighbor circuit of depth D, the backward causal cone of a single-site observable contains at most $\mathcal{O}(D)$ qubits. If each layer contains $\mathcal{O}(n)$ parameters, then the total number of parameters is typically $p = \mathcal{O}(nD)$, while the number of parameters inside the causal cone of one local observable is at most $\mathcal{O}(D^2)$. Therefore, for fixed or slowly growing D,

$$k = \mathcal{O}(D^2), \qquad p = \mathcal{O}(nD), \qquad \frac{k}{p} \to 0 \quad \text{as } n \to \infty. \tag{40}$$

This gives an explicit setting in which locality and shallow depth imply exact sparse parameter gradients. This causal-cone viewpoint is consistent with analyses showing that the gradient behavior of local Pauli cost terms depends on the width of their causal cones.

1.2.2 Algebraic Commutation-Induced Exact Sparsity

The causal-cone analysis first separates parameters into two groups: parameters outside the relevant backward causal cones and parameters inside them. Parameters outside the cones are inactive by locality. However, being inside a causal cone is only a necessary condition for a parameter to affect the objective; it is not sufficient. Even if a parameterized gate lies inside the causal cone, its gradient can still be zero if its generator commutes with the corresponding effective observable.

For a parameter inside the causal active set $\mathcal{B}$, the gradient is still governed by

$$\partial_j C(\boldsymbol{\theta}) = \frac{i}{2} \operatorname{Tr}\left[\rho_j(\boldsymbol{\theta})[G_j, O_j(\boldsymbol{\theta})]\right]. \tag{41}$$

Therefore, if

$$[G_j, O_j(\boldsymbol{\theta})] = 0, \tag{42}$$

then

$$\partial_j C(\boldsymbol{\theta}) = 0. \tag{43}$$

This is an algebraic mechanism. Unlike causal-cone sparsity, the generator G_j and the effective observable O_j may act on the same qubits. The gradient vanishes because the parameterized direction is a commuting, symmetry-preserving direction of the objective.

Define the algebraically active set

$$\mathcal{A} = \{j \in \mathcal{B} : [G_j, O_j(\boldsymbol{\theta})] \neq 0\}. \tag{44}$$

Then

$$\partial_j C(\boldsymbol{\theta}) = 0, \qquad \forall j \notin \mathcal{A}. \tag{45}$$

Thus, even if the causal active set $\mathcal{B}$ is large, the gradient can still be sparse when the noncommuting set $\mathcal{A}$ is small. If

$$|\mathcal{A}| = k, \qquad k = o(p), \tag{46}$$

then the parameter gradient is exactly sparse.

This mechanism is naturally connected to Lie-algebraic analyses of VQCs. For circuits generated by Pauli strings or Hamiltonian terms, gradients can be expressed through commutators and nested commutators in the associated dynamical Lie algebra. Consequently, only directions that are algebraically connected to the measured observable can contribute to the gradient, while generators in the commutant of the observable produce zero-gradient directions.

Concrete commutation example. Consider an Ising or MaxCut-type Hamiltonian

$$O_C = \sum_{(u,v)\in E} w_{uv} Z_u Z_v, \tag{47}$$

which is widely used in QAOA and multi-angle QAOA. Let the ansatz contain a mixer block followed by a terminal diagonal phase block:

$$U(\boldsymbol{\beta}, \boldsymbol{\gamma}) = U_D(\boldsymbol{\gamma}) U_M(\boldsymbol{\beta}), \tag{48}$$

where

$$U_D(\boldsymbol{\gamma}) = \prod_{(u,v)\in E} e^{-i\gamma_{uv} Z_u Z_v/2}, \tag{49}$$

and

$$U_M(\boldsymbol{\beta}) = \prod_{i=1}^{r} e^{-i\beta_i X_i/2}. \tag{50}$$

The generators of the diagonal block are the Pauli strings $Z_u Z_v$. They form an Abelian Lie subalgebra and commute with the measured Hamiltonian:

$$[Z_u Z_v, O_C] = 0, \qquad \forall (u,v) \in E. \tag{51}$$

Therefore,

$$U_D^\dagger(\boldsymbol{\gamma}) O_C U_D(\boldsymbol{\gamma}) = O_C. \tag{52}$$

Using this relation in the objective function gives

$$C(\boldsymbol{\beta}, \boldsymbol{\gamma}) = \mathrm{Tr}\left[O_C U_M(\boldsymbol{\beta}) \rho_0 U_M^\dagger(\boldsymbol{\beta}) \right]. \tag{53}$$

Hence the objective is independent of all diagonal parameters γ_{uv}, and

$$\frac{\partial C}{\partial \gamma_{uv}} = 0, \qquad \forall (u,v) \in E. \tag{54}$$

In contrast, mixer generators generally do not commute with O_C. If qubit i is incident to an edge $(i,j) \in E$, then

$$[X_i, Z_i Z_j] \neq 0. \tag{55}$$

Thus, only the mixer parameters are potentially active.

The total number of parameters is

$$p = |E| + r, \tag{56}$$

where $|E|$ parameters correspond to commuting diagonal generators and r parameters correspond to potentially noncommuting mixer generators. The number of potentially active parameters satisfies

$$k \leq r. \tag{57}$$

For a dense graph with $|E| = \Theta(n^2)$ and a mixer acting on $r = n$ qubits,

$$p = \Theta(n^2), \qquad k \leq n, \qquad \frac{k}{p} = \mathcal{O}\left(\frac{1}{n}\right) \to 0. \tag{58}$$

Therefore, this ansatz-objective pair gives an explicit example of exact sparse parameter gradients.

In summary, causal-cone sparsity and algebraic commutation sparsity are complementary. Causal cones identify which parameters can structurally influence a local observable. Among those parameters, commutation relations further determine which directions are algebraically active. A circuit satisfies exact sparse parameter gradients when the final active set, after both structural and algebraic filtering, has size $k = o(p)$.

1.2.3 Observable-Tail-Induced Approximate Sparse Gradients

The previous two mechanisms give exact zero-gradient components. In this subsection, we discuss a softer but physically relevant mechanism: approximate sparsity induced by decaying Hamiltonian tails. This case is important for long-range many-body systems, where distant interaction terms are not exactly absent but their coefficients or operator norms decay with distance.

Consider a VQE-type objective in which the measured observable O is the Hamiltonian:

$$C(\boldsymbol{\theta}) = \mathrm{Tr}\left[OU(\boldsymbol{\theta})\rho_0 U^\dagger(\boldsymbol{\theta})\right]. \tag{59}$$

Assume that O admits a decomposition into local or quasi-local terms:

$$O = \sum_X O_X = \sum_X J_X h_X, \tag{60}$$

where X denotes a subset of lattice sites or qubits, h_X is an operator supported on X, and J_X is its coupling coefficient. Then

$$C(\boldsymbol{\theta}) = \sum_X J_X \langle h_X \rangle_{\boldsymbol{\theta}}, \tag{61}$$

where

$$\langle h_X \rangle_{\boldsymbol{\theta}} = \mathrm{Tr}\left[h_X U(\boldsymbol{\theta})\rho_0 U^\dagger(\boldsymbol{\theta})\right]. \tag{62}$$

Therefore, the j-th gradient component is

$$\partial_j C(\boldsymbol{\theta}) = \sum_X J_X \partial_j \langle h_X \rangle_{\boldsymbol{\theta}}. \tag{63}$$

Let the parameter θ_j appear in a local gate

$$U_j(\theta_j) = e^{-i\theta_j G_j/2}, \tag{64}$$

where the generator G_j is supported on a small region S_j. In a local shallow circuit of depth D, the influence of this parameter can spread only to an effective neighborhood of S_j, denoted by

$$\mathcal{N}_D(S_j). \tag{65}$$

Thus, a Hamiltonian term h_X can have a large sensitivity to θ_j only if its support X overlaps or is close to this effective neighborhood.

For a cutoff radius R, decompose the gradient into a local part and a tail part:

$$\partial_j C(\boldsymbol{\theta}) = \mathrm{local}_j(R) + \mathrm{tail}_j(R), \tag{66}$$

where

$$\mathrm{local}_j(R) = \sum_{\mathrm{dist}(X,S_j)\leq R} J_X \partial_j \langle h_X \rangle_{\boldsymbol{\theta}}, \tag{67}$$

and

$$\mathrm{tail}_j(R) = \sum_{\mathrm{dist}(X,S_j)> R} J_X \partial_j \langle h_X \rangle_{\boldsymbol{\theta}}. \tag{68}$$

If the termwise sensitivity is bounded as

$$|\partial_j \langle h_X \rangle_{\boldsymbol{\theta}}| \leq C_h, \tag{69}$$

then

$$|\mathrm{tail}_j(R)| \leq C_h \sum_{\mathrm{dist}(X,S_j)> R} |J_X|. \tag{70}$$

For finite-range Hamiltonians, sufficiently distant terms are exactly absent, so the tail contribution vanishes beyond the interaction range. For long-range Hamiltonians with decaying interactions, the tail is not zero but can be controlled. A common long-range assumption is that the total strength of terms with large diameter decays as a power law:

$$f(R) = \sup_z \sum_{\substack{Z\ni z \\ \mathrm{diam}(Z)\geq R}} \|O_Z\| \leq \frac{J}{R^\alpha}, \qquad \alpha > d, \tag{71}$$

where d is the spatial dimension of the lattice. This type of assumption appears in Lieb–Robinson bounds for long-range interacting systems. It means that, although long-range terms exist, their total strength around any fixed site is controlled by a decaying tail.

Under such a decay condition, Eq. (70) implies that

$$|\mathrm{tail}_j(R)| \leq \frac{C'}{R^\alpha} \tag{72}$$

for some constant C'. Hence,

$$\partial_j C(\boldsymbol{\theta}) = \mathrm{local}_j(R) + \mathcal{O}(R^{-\alpha}). \tag{73}$$

This shows that each parameter gradient is mainly determined by nearby Hamiltonian terms, while distant terms contribute only a controlled small correction.

This mechanism does not automatically imply that the whole gradient vector is sparse. It implies sparsity only when the large local contributions are concentrated around a small number of parameters. Define the local interaction strength seen by parameter θ_j as

$$W_j(R) = \sum_{\mathrm{dist}(X,S_j)\le R} |J_X|. \tag{74}$$

Parameters with large $W_j(R)$ are close to important Hamiltonian terms and can have large gradients. Parameters with small $W_j(R)$ interact mainly with weak local terms and decaying tails, so their gradients are small.

For a threshold $\lambda > 0$, define the approximately active set

$$\mathcal{A}_{R,\lambda} = \{j \in \{1,\dots,p\} : W_j(R) > \lambda\}. \tag{75}$$

If

$$|\mathcal{A}_{R,\lambda}| = k, \qquad k = o(p), \tag{76}$$

and for all $j \notin \mathcal{A}_{R,\lambda}$,

$$|\partial_j C(\boldsymbol{\theta})| \le C_h \lambda + \frac{C'}{R^\alpha}, \tag{77}$$

then most gradient components are small. If the top-k components capture a fraction ρ of the total squared gradient energy,

$$\frac{\sum_{j\in S_k(\boldsymbol{\theta})} \left(\partial_j C(\boldsymbol{\theta})\right)^2}{\sum_{j=1}^{p} \left(\partial_j C(\boldsymbol{\theta})\right)^2} \ge \rho, \tag{78}$$

then the gradient satisfies the (k,ρ)-sparse condition. In contrast to the causal-cone and commutation mechanisms, this is generally an approximate sparsity or compressibility result rather than exact sparsity.

Example: one-dimensional spin chain with decaying tails. Consider an n-qubit spin chain with Hamiltonian

$$O = O_{\mathrm{local}} + O_{\mathrm{tail}}, \tag{79}$$

where

$$O_{\mathrm{local}} = \sum_{i=1}^{n-1} J_i Z_i Z_{i+1}, \tag{80}$$

and

$$O_{\mathrm{tail}} = \sum_{1\le i<j\le n} J_{ij} Z_i Z_j. \tag{81}$$

Assume that the long-range couplings decay as

$$|J_{ij}| \le \frac{J_0}{|i-j|^\alpha}, \qquad \alpha > 1. \tag{82}$$

Let the ansatz be a local shallow circuit consisting of L layers of single-qubit rotations and nearest-neighbor entangling gates. Suppose each layer contains one trainable rotation per qubit, so that

$$p = nL. \tag{83}$$

A parameter θ_j acting near qubit s_j can mainly affect terms supported inside a light-cone neighborhood of s_j. If the circuit depth is D, this neighborhood has size at most $\mathcal{O}(D)$.

Now suppose the dominant local couplings are concentrated in a small spatial region Ω of size m, while the remaining terms are weak or belong to the decaying tail. Then only parameters whose light cones intersect Ω can receive large local-gradient contributions. The number of such parameters is bounded by

$$k = \mathcal{O}(L(m + D)). \tag{84}$$

Since the total number of parameters is $p = nL$, if m and D are fixed or grow much more slowly than n, then

$$\frac{k}{p} = \mathcal{O}\left(\frac{m + D}{n}\right) \to 0. \tag{85}$$

For parameters outside this active region, the local contribution is small and the remaining long-range contribution is suppressed by the decay of J_{ij}. Therefore, their gradient components are small rather than exactly zero.

This gives a concrete approximate-sparsity scenario:

- parameters whose light cones overlap strong local Hamiltonian terms have large gradients;
- parameters that only interact with weak or far-away tail terms have small gradients;
- if the number of parameters near the strong terms is $k = o(p)$, then the gradient vector is compressible and can satisfy the (k, ρ)-sparse condition for ρ close to one.

Thus, observable-tail decay provides a physically motivated example of approximate sparse parameter gradients. Finite-range Hamiltonians can give exact locality in the parameter-term Jacobian, while long-range Hamiltonians with decaying interactions give small but nonzero tail contributions. This is precisely the regime where Select-SPSA is useful: the dominant optimization signal is concentrated in a small active subset of parameters, while the remaining components contribute only weakly to the total gradient energy.

Sparse parameter gradients can arise naturally in variational quantum circuits. We discuss two common sources: sparsity induced by the circuit architecture and sparsity induced by the properties of the operators.

2 algorithm

Here, $\hat{\mathbf{g}}_t^{\mathrm{SPSA}}$ denotes the full SPSA gradient estimate, $s_{t,j}$ is the selection score of parameter θ_j, S_t is the selected top-k parameter set, and $\hat{\mathbf{g}}_t^{\mathrm{SG}}$ is the final hybrid gradient used for the parameter update.

Algorithm 1 Select-Guided-SPSA

Require: Objective function $C(\boldsymbol{\theta})$, initial parameters $\boldsymbol{\theta}_0 \in \mathbb{R}^p$, learning rate η_t, SPSA perturbation scale c_t, number of selected parameters k, number of iterations T.

Ensure: Optimized parameters $\boldsymbol{\theta}_T$.

for $t = 0, 1, \ldots, T-1$ **do**

Sample a random perturbation vector $\boldsymbol{\Delta}_t \in \{-1, 1\}^p$.

Compute the SPSA gradient estimate for all parameters:

$$\hat{\mathbf{g}}_t^{\mathrm{SPSA}} = \frac{C(\boldsymbol{\theta}_t + c_t \boldsymbol{\Delta}_t) - C(\boldsymbol{\theta}_t - c_t \boldsymbol{\Delta}_t)}{2c_t} \boldsymbol{\Delta}_t^{-1}.$$

Score each parameter by the magnitude of its SPSA estimate:

$$s_{t,j} = \left| \hat{g}_{t,j}^{\mathrm{SPSA}} \right|, \qquad j = 1, \ldots, p.$$

Select the top-k parameters:

$$S_t = \mathrm{TopK}\left(\{s_{t,j}\}_{j=1}^{p}, k \right).$$

Initialize the hybrid gradient estimate:

$$\hat{\mathbf{g}}_t^{\mathrm{SG}} \leftarrow \hat{\mathbf{g}}_t^{\mathrm{SPSA}}.$$

for each $j \in S_t$ **do**

Recompute the selected gradient component by the parameter-shift rule:

$$\hat{g}_{t,j}^{\mathrm{PSR}} = \frac{1}{2} \left[C\left(\boldsymbol{\theta}_t + \frac{\pi}{2} \mathbf{e}_j \right) - C\left(\boldsymbol{\theta}_t - \frac{\pi}{2} \mathbf{e}_j \right) \right].$$

Replace the SPSA estimate on the selected coordinate:

$$\hat{g}_{t,j}^{\mathrm{SG}} \leftarrow \hat{g}_{t,j}^{\mathrm{PSR}}.$$

end for

Update the parameters:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \hat{\mathbf{g}}_t^{\mathrm{SG}}.$$

end for

return $\boldsymbol{\theta}_T$.

3 Experiments

3.1 Toy problem

We first evaluate Select-Guided-SPSA on a one-dimensional toy problem. The purpose of this experiment is to construct a simple nonconvex landscape in which a purely local gradient-following method is attracted to a local minimum, while Select-Guided-SPSA can reach the basin of the global minimum through stochastic SPSA-guided exploration.

The objective function is the quartic double-well function

$$L(x) = \frac{1}{4}x^4 + 0.227924077994387\,x^3 + 0.0251316701949487\,x^2. \tag{86}$$

Its derivative factorizes as

$$L'(x) = x(x + 0.6)(x + 0.0837722339831621). \tag{87}$$

Therefore, the function has a local minimum at $x = 0$, a global minimum near $x = -0.6$, and a separating barrier near $x = -0.0838$. The initial point is set to

$$x_0 = 0.05, \tag{88}$$

which lies in the basin of attraction of the local minimum for a local gradient-based method.

We compare the parameter-shift rule (PSR) optimizer with Select-Guided-SPSA from the same initialization. PSR follows the local gradient direction and is therefore attracted to the nearby local minimum at $x = 0$. In contrast, Select-Guided-SPSA uses stochastic SPSA perturbations to screen promising directions and can visit the global basin near $x = -0.6$.

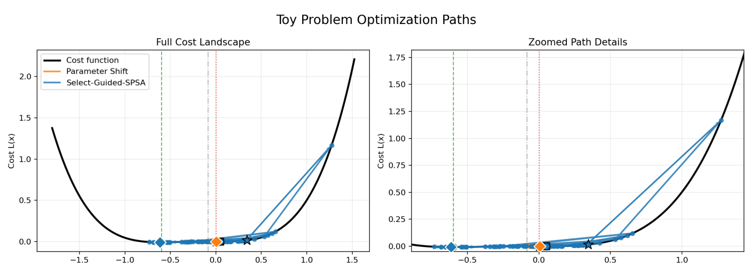

Figure 2: Optimization trajectories on the toy double-well objective. Parameter-shift gradient descent converges to the local minimum near $x = 0$, whereas Select-Guided-SPSA follows a stochastic trajectory that escapes the local basin and reaches the global-minimum basin near $x = -0.6$. The right panel provides a magnified view of the optimization paths near the two minima.

The result in Figure 2 shows the intended behavior of the toy construction. The PSR trajectory remains trapped in the local well near $x = 0$. Select-Guided-SPSA, however, discovers a better point in the global basin. This experiment does not require monotonic convergence of the stochastic trajectory; instead, it demonstrates that the SPSA-guided selection mechanism can help the optimizer access a basin that is difficult for a strictly local gradient-following method to reach from the same initialization.

3.2 Quantum Circuit Examples Exhibiting Parameter-Gradient Sparsity

3.2.1 example 1

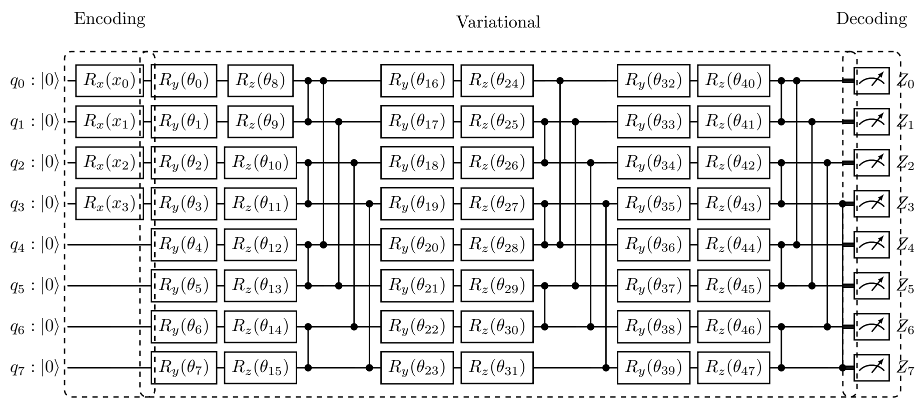

Figure 3: Sparse-local variational quantum circuit for Friedman2. The first four qubits encode the input features using $R_x(x_j)$, while q_4–q_7 are auxiliary qubits without direct input encoding. Each variational layer applies trainable R_y and R_z rotations to all qubits, followed by alternating local CZ gates and Friedman2 feature-pair CZ gates $(0,4),(1,5),(2,6),(3,7)$. The decoder measures the local observables $Z_0,\ldots,Z_7$.

Table 1: PSR gradient-energy sparsity and regression performance on Friedman2. M denotes the number of trainable quantum parameters. Coverage is the cumulative squared-gradient energy captured by the top-k parameters.

Circuit / run	Stage	M	k	k/M	gradient coverage	Test metric
Sparse-local psr	epoch 5	48	3	0.0625	0.9073	0.3856
Sparse-local psr	epoch 10	48	7	0.1458	0.9231	
Sparse-local psr	epoch 15	48	7	0.1458	0.9180	
Sparse-local psr	epoch 20	48	3	0.0625	0.9009	
sparse-local s-g-s		48	7	0.1458	–	0.3686

The sparse-local ansatz exhibits a strongly concentrated PSR gradient-energy profile: only three out of forty-eight parameters are sufficient to explain approximately 90% of the loss-gradient energy at several training checkpoints. This confirms that the parameter-gradient vector is highly sparse under the PSR estimator. However, the same circuit obtains a substantially worse test metric than the less sparse Skolik baseline. Therefore, PSR gradient sparsity is not by itself a sufficient condition for good trainability or predictive performance.

A plausible explanation is that the sparse-local design imposes a strong locality bias on the variational hypothesis class. The circuit encodes the four Friedman2 features on separate feature qubits and couples them mainly through alternating nearest-neighbor CZ gates and fixed feature-pair links. This restricts the number of paths through which information can propagate between distant qubits. As a result, the measured observables can become sensitive to only a small subset of rotations, producing a sparse gradient-energy profile. At the same time, the limited entanglement connectivity can reduce the effective Fourier spectrum and suppress high-order correlation terms in the model output. Since Friedman2 contains nonlinear feature interactions, weak global mixing can limit the circuit's ability to represent the target function.

From this viewpoint, the observed sparsity is a double-edged property. It makes the dominant parameters easy to identify, but it may also indicate that many parameters are functionally inactive for the chosen observables. The Skolik baseline has a denser ring-entangling structure and a less concentrated gradient profile, requiring about 20%–25% of the parameters to reach 80% gradient-energy coverage. Although its gradients are less sparse, the stronger information propagation gives it better expressive capacity and leads to a much lower test error. This suggests that Select-SPSA circuit design should balance gradient sparsity with sufficient entangling expressivity, instead of maximizing sparsity alone.

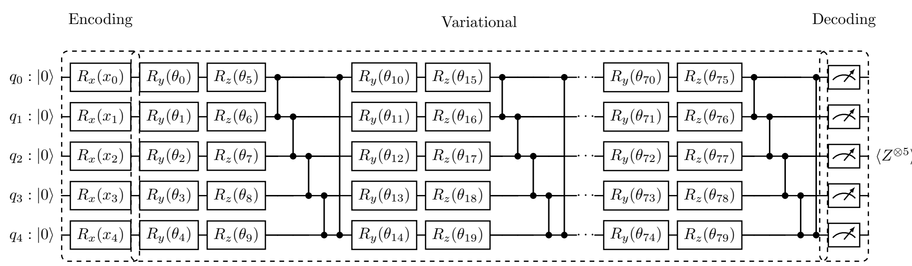

Figure 4: Compact representation of the `skolik_arch` variational quantum circuit used in the 80-parameter Select-Guided-SPSA setting. The encoding block applies $R_x(x_j)$ to the five input features. The variational block contains eight repeated layers; each layer applies trainable R_y and R_z rotations on all five qubits, followed by a ring of CZ gates $(0, 1), (1, 2), (2, 3), (3, 4), (4, 0)$. Layers 1, 2, and 8 are shown explicitly. The middle ellipsis represents layers 3–7, which have the same rotation-plus-ring-CZ structure and contain parameters $\theta_{20}, \ldots, \theta_{69}$. The decoding block evaluates the global Pauli observable $Z^{\otimes 5} = Z_0 Z_1 Z_2 Z_3 Z_4$, so the circuit output is the scalar expectation value $\langle Z^{\otimes 5} \rangle$, not five separate single-qubit observables.

3.3 Gradient Sparsity and Performance on Friedman1

We evaluate the proposed selection-based gradient strategy on the Friedman1 regression task using the Skolik-type variational circuit shown in Fig. 4. The

Table 2: PSR gradient sparsity on Friedman1 with the 80-parameter Skolik circuit in the ideal setting. The probe is performed every five epochs with target energy ratio $\rho = 0.8$.

Epoch	k	k/M	Coverage
5	25	0.3125	0.8078
10	29	0.3625	0.8069
15	22	0.2750	0.8030
20	26	0.3250	0.8082

circuit contains $n = 5$ qubits and $L = 8$ variational layers. Each layer applies one R_y and one R_z trainable rotation to every qubit, followed by a ring of controlled-Z gates. Therefore, the number of trainable quantum parameters is

$$M = 2nL = 2 \times 5 \times 8 = 80. \tag{89}$$

We use a training set size of 100, batch size 20, and train for 20 epochs. Thus, each epoch contains 5 training batches.

Gradient sparsity under exact PSR. To examine whether the parameter-shift-rule (PSR) gradients are sparse, we compute the exact PSR gradient during training and perform a sparsity probe every five epochs. For a target energy ratio $\rho = 0.8$, we sort the parameters by the squared gradient magnitude and report the smallest number of parameters k whose cumulative gradient energy reaches at least 80% of the full gradient energy:

$$\text{coverage}(k) = \frac{\sum_{i \in \mathcal{S}_k} g_i^2}{\sum_{j=1}^{M} g_j^2}, \tag{90}$$

where g_i denotes the PSR gradient of the i-th trainable parameter, and $\mathcal{S}_k$ is the set of the top-k parameters ranked by g_i^2. The value "coverage" therefore measures how much of the full gradient energy is captured by the selected parameters.

The results in Table 2 show that only about 27.5%–36.25% of the parameters are needed to capture approximately 80% of the PSR gradient energy. This confirms that the exact gradient exhibits a clear sparsity structure in this setting, which supports the use of a selective PSR refinement mechanism.

Circuit-count comparison. Let B be the batch size, M the number of trainable parameters, m the SPSA batch size, k the number of selected parameters, and T the total number of training batches. In our setting,

$$B = 20, \quad M = 80, \quad T = 20 \times 5 = 100. \tag{91}$$

Table 3: Performance comparison on Friedman1 with the 80-parameter Skolik circuit in the ideal setting. Lower test metric is better.

Method	Test Metric	Convergence Epoch	Total Circuits
PSR	0.0964	18	320000
SPSA-10	0.1526	14	40000
SPSA-21	0.1362	18	84000
Select-Guided-SPSA-10, top-10	0.1259	14	80000

Table 4: Performance comparison on Friedman1 with the 80-parameter Skolik circuit in the shot-based setting. Lower test metric is better.

Method	Test Metric	Total Circuits
SPSA-10	0.2291	40000
SPSA-21		84000
PSR	0.1566	320000
Select-Guided-SPSA, top-11	0.1578	84000

The gradient circuit counts are computed as

$$C_{\text{PSR}} = T \cdot 2BM, \tag{92}$$

$$C_{\text{SPSA}} = T \cdot 2Bm, \tag{93}$$

$$C_{\text{Select}} = T \cdot (2Bm + 2Bk)\,. \tag{94}$$

Here, PSR requires two shifted circuits per parameter and sample, SPSA uses m random perturbation directions, and Select-Guided-SPSA first estimates the gradient using SPSA and then refines the selected top-k parameters using PSR.

Ideal simulation results. Table 3 compares PSR, SPSA, and Select-Guided-SPSA under the ideal analytical simulator.

The exact PSR method achieves the best test metric, but it requires the largest number of gradient circuits. Compared with SPSA-10, Select-Guided-SPSA improves the test metric from 0.1526 to 0.1259 while using twice the number of circuits. Compared with SPSA-21, which uses a similar circuit budget, Select-Guided-SPSA achieves a better test metric, suggesting that selectively refining important parameters with PSR is more effective than simply increasing the number of SPSA perturbation directions.

Shot-based simulation results. For completeness, Table 4 reports the corresponding shot-based results on the same 80-parameter circuit.

The shot-based results show a similar trend. Select-Guided-SPSA achieves a test metric close to PSR while using substantially fewer circuits. In particular, it reduces the circuit cost from 320000 to 84000, corresponding to about 26.25% of the PSR circuit budget, while maintaining comparable predictive performance.

References

[1] D. V. Else, F. Machado, C. Nayak, and N. Y. Yao, "An improved Lieb-Robinson bound for many-body Hamiltonians with power-law interactions," *Physical Review A*, vol. 101, no. 2, p. 022333, 2020, doi: 10.1103/PhysRevA.101.022333.